

Automation Risk, Redistribution, and Economic Vulnerability: Evidence from the European Social Survey, 2002–2022

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Abstract

This paper asks whether occupational exposure to automation is associated with support for government redistribution, and when that association is strongest. We reconstruct and extend the research design of Thewissen and Rueda [2019] with all eleven European Social Survey (ESS) waves from 2002 to 2022, covering more than 275,000 respondents in 39 countries. The positive association between automation risk and redistribution preferences reappears in the 2002–2012 period but is smaller in the subsequent decade. Country-by-year estimates show substantial variation across national settings. We then examine two mechanism probes. Automation exposure is associated with perceived economic vulnerability, including financial strain and job insecurity. Institutions and industrial relations condition the association between exposure and felt insecurity: after adjustment for business-cycle dynamics, union density weakens it within and between countries, whereas net income inequality and strict employment protection strengthen it. The latter pattern is consistent with insider–outsider dualization.

1 Introduction

Automation and artificial intelligence have renewed debate over the political consequences of labor-market displacement. Comparative political economists use public-opinion data to study how economic vulnerability shapes policy preferences. In a central contribution, Thewissen and Rueda [2019] found that respondents in occupations with greater routine-task automation risk expressed more support for redistribution. This paper asks whether that association persists across periods and national contexts.

We rebuild the Thewissen–Rueda design from raw ESS records harmonized to a common format for eleven waves between 2002 and 2022. First, we extend the temporal coverage beyond 2012, when cognitive automation and artificial intelligence became more salient. Second, we estimate country and country-year slopes to assess whether pooled estimates conceal geographic variation. Third, we examine whether perceived vulnerability links occupational exposure to redistribution preferences and whether welfare-state and labor-market institutions condition that link.

Methodologically, the empirical literature operationalizes automation risk through occupational exposure measures, most often the routine-task and engineering-bottleneck probability of Frey and Osborne [2017] or the routine-task-intensity index of Goos et al. [2014]. Both measures capture the same underlying idea — that codifiable, repetitive tasks are the ones most readily displaced by digitization and machine automation. We use both throughout, treating the Frey–Osborne probability as the reference measure and the routine-task-intensity index as the closest robustness analogue to the Thewissen–Rueda benchmark.

The focus remains redistribution preferences. The vulnerability and institutional analyses clarify when occupational exposure translates into those preferences.

We first report the Frey–Osborne replication of the Thewissen–Rueda design and its post-2014 attenuation. We then examine country-year slopes and composition-adjusted yearly estimates. The final section presents the vulnerability and institutional analyses; routine-task-intensity (RTI) results and additional robustness checks appear in the appendix.

2 Theory

2.1 Comparative Political Economy of Redistribution

Our framework draws on two accounts of welfare-state development and redistribution. Wilensky's modernization account treats welfare-state expansion as a response to industrialization, economic growth, and demographic change [Wilensky, 1974]. Stephens' power-resources account emphasizes labor organization, socialist parties, and collective bargaining as forces that sustain redistribution [Stephens, 1979]. Together, these accounts distinguish the creation of economic risk from its political consequences: institutions and class power shape how people experience new risks.

The comparative political economy literature also clarifies why redistribution demand need not mechanically follow from objective exposure. Moene and Wallerstein [2001] show that preferences over social insurance and redistribution depend on the distribution of risk and income, not on inequality alone. Iversen and Soskice's asset theory similarly treats social-policy preferences as a function of exposure to labor-market risk and the portability of workers' skills [Iversen and Soskice, 2001]. Korpi and Palme's paradox of redistribution adds an institutional feedback: encompassing welfare states can reduce poverty and inequality precisely because broad social programs build larger political coalitions for redistribution [Korpi and Palme, 1998]. Together these arguments imply that automation exposure should matter politically when it is experienced as uninsured labor-market risk, and that welfare-state and labor-market institutions should condition that experience.

2.2 Job Automation and Economic Threat

Automation can alter perceptions of economic threat, which in turn can shape political attitudes. Research on economic shocks links perceived insecurity, especially anticipated unemployment, to changes in political and social preferences [Margalit, 2019, 2013]. Research on economic vulnerability and political psychology likewise treats perceived threats as a potential source of support for protective policies [Hacker et al., 2013, Rogers, 1975]. Automation is one poten-

tial source of such disruption: people in exposed occupations may regard their employment prospects as less secure [Autor and Salomons, 2018].

2.3 Automation Risk as Conditional Redistribution Demand

The benchmark claim is that routine-task automation risk increases demand for redistribution because occupational exposure can signal future income and employment losses [Thewissen and Rueda, 2019]. Respondents in routine, automatable occupations may anticipate displacement or downward mobility, making redistribution and social insurance more attractive. We therefore expect automation exposure to be positively associated, on average, with support for redistribution.

The association need not be constant over time. Economic development, employment structures, and institutions change, altering how occupational risk is understood. Recessions, recoveries, and inflation can make the same exposure feel more or less threatening. People and firms also adapt as automation diffuses, through changes in expectations, tasks, and occupations. Routine-task measures may therefore lose political relevance as technology moves beyond routine-biased computerization. We expect the automation–redistribution association to vary across survey waves rather than to move monotonically in one direction.

The same exposure may also have different consequences across countries. Welfare states and labor-market institutions shape whether automation risk is privately borne or collectively insured. Strong labor organization and broad bargaining can provide leverage before market risk becomes individualized. Generous unemployment protection, social spending, and redistribution can buffer the material and psychological consequences of exposure. Where such protection is weaker or inequality is higher, exposure may generate greater demand for redistribution because the risk is less insured.

We use perceived vulnerability to trace how exposure may translate into redistribution demand. Exposure may appear first in expectations of job loss, financial strain, and insufficient income. We therefore advance three hypotheses: H_1 expects higher occupational automation risk to predict more support for redistribution; H_2 expects that association to vary across time; and H_3 expects it to vary across countries. We also use two mechanism probes. M_1 asks whether

perceived economic vulnerability partly links automation risk to redistribution demand. M_2 asks whether welfare-state and labor-market institutions moderate the association between exposure, vulnerability, and redistribution.

2.4 Measuring Routine-Task Automation Risk

The measures used in this literature rate occupations by the routineness of their task content, because computers and machines can more readily perform codifiable, repetitive tasks [Autor and Salomons, 2018]. We use the engineering-bottleneck probability of Frey and Osborne [2017] and the routine-task-intensity (RTI) index of Goos et al. [2014]; they correlate at 0.573 in our sample. Both measures are highest in clerical, production, and lower-skill service occupations, where pay and employment security also tend to be lower.

This is the construct used in the redistribution literature, and we retain it throughout the paper. Frey–Osborne is the reference measure; RTI is the robustness measure. If perceived vulnerability tracks labor-market precarity, routine-task exposure should predict greater vulnerability among respondents in the occupations it marks. The automation–redistribution association should be strongest where institutions offer the least protection against that vulnerability.

3 Data and Measures

The analysis uses a harmonized panel of eleven European Social Survey (ESS) waves from 2002 to 2022, spanning 39 countries and roughly 275,000 respondents before listwise deletion. The outcome is the standard ESS item asking whether government should reduce income differences. It is coded on a five-point scale, with higher values indicating greater support for redistribution. Following Thewissen and Rueda [2019], the main analysis is limited to employed and unemployed members of the active labor force.

Figure 1: Distribution of Job Automation Risks in the ESS Sample

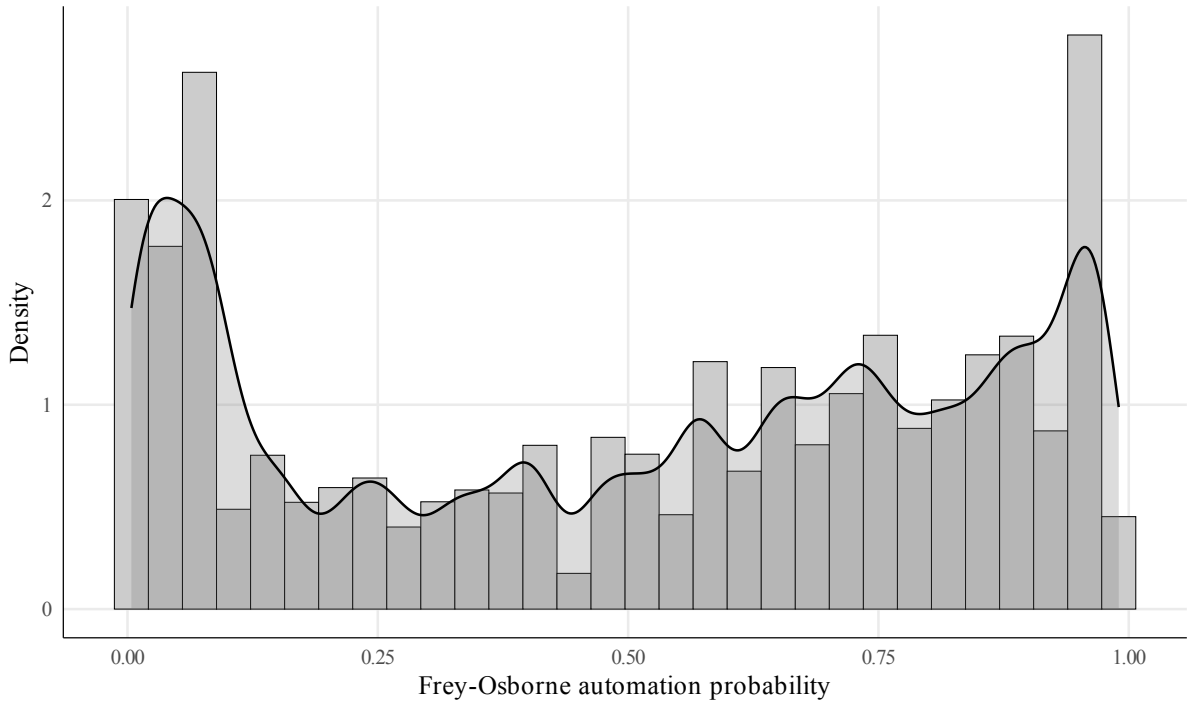


Figure 1 shows Frey–Osborne automation probabilities for ESS occupations.¹ Because this measure is a probability, observations cluster toward the ends of the 0–1 scale. Its density resembles the corresponding U.S. distribution in Frey and Osborne [2017], which is consistent with a successful occupation match.

Frey–Osborne is our reference index, and alongside it we use the Goos–Manning–Salomons routine-task-intensity (RTI) index as a second, independently constructed measure of the same routine-task construct. The panel bridges the ESS occupation-coding seam: rounds 1–5 use ISCO-88 and rounds 6–11 use ISCO-08, so we harmonize both exposure measures across the two occupation systems.

We measure economic position with income that is equivalized for household size and expressed relative to the country-year mean. We convert income-band responses in ESS show cards to monetary midpoints. For the open top band, we estimate a Pareto mean from the observed counts in the top two categories, following Hout [2004]. We then apply the square-root equivalence scale and divide the resulting value by the design-weighted country-year mean. This control matches the baseline specification in the benchmark literature.

¹The match covers almost 90% of potential cases; matched and unmatched cases do not differ systematically.

We validated the harmonization by joining the raw panel to the original replication data of Thewissen and Rueda [2019] at the respondent level for rounds 1–6. RTI matches exactly, Frey–Osborne probabilities match almost exactly, and the redistribution outcome and core demographic variables have near-perfect agreement. These checks support treating the 2002–2022 panel as a direct extension of the published benchmark design.

The mechanism analyses use several indicators of economic vulnerability. Financial strain is available in every wave and is coded as whether the household finds it difficult or very difficult to cope on its current income. Selected waves provide additional outcomes: expected job loss and expected income insufficiency in the 2008 and 2016 welfare-attitudes modules, and perceived job insecurity in the 2004 and 2010 work modules. All are restricted to labor-market participants.

At the country level, we examine ten institutional moderators. The Comparative Political Data Set supplies six annual series [CPDS; Armingeon et al., 2025]: union density, collective-bargaining coverage, employment-protection strictness, left cabinet share, industry employment share, and trade openness. Four additional series come from their source APIs: public social spending as a share of GDP (OECD SOCX), the replacement rate for net unemployment benefits (OECD tax–benefit models), net income inequality, and relative redistribution (SWIID, version 9.92). The two-stage diagnostic uses their 2016 values. The moderation models use country-year values that are standardized and lagged by one year. Depending on availability, those models contain 20–39 national clusters.

For redistribution preferences, we estimate one multilevel model with random effects for each exposure index. It regresses individual support for redistribution on occupational automation exposure and the benchmark controls. The automation-risk slope has global, country, year, and country-year components (equations 1–2). Every reported redistribution result is derived from that fit. The country and year profiles are random-effect components; the country-year map is the composite $\beta_{c,t}$; and the replication, composition, and institutional diagnostics aggregate or correlate those fitted slopes. We do not re-estimate redistribution coefficients on subsamples.

To ensure consistency with the published benchmark, our primary models include the baseline control vector established by Thewissen and Rueda [2019]. This vector controls for relative

equivalized income, years of education, gender, age, current or former union membership, religiosity, and individual unemployment status, adjusted using ESS design weights and accounting for the country-year structure of the survey. The appendix reports sensitivity checks with household size, partner status, and self-placement on the left-right scale.

We estimate the mechanism analyses separately from the redistribution models. At the individual level, mediation models treat financial strain and job insecurity as intervening variables between occupational exposure and support for redistribution. At the country level, each institutional measure is interacted separately with individual automation risk. Models using multiple waves include one-year-lagged moderators, year fixed effects, exposure slopes that vary by country, and a country-year random intercept. The intercept absorbs wave-level macro shocks that would otherwise understate the standard errors of interactions across levels in pooled cross-sections [Schmidt-Catran and Fairbrother, 2016]. These models assess whether protective institutions weaken the association between occupational risk and perceived vulnerability.

Let y_{ict} denote the outcome for individual i in country c in survey year t , and let AR_{ict} denote the occupational automation-risk index. The redistribution model is:

$$y_{ict} = \alpha_{ct} + \beta_{ct}AR_{ict} + X_{ict}\Gamma + \varepsilon_{ict}, \quad (1)$$

where α_{ct} is a country-year-specific intercept, X_{ict} is the vector of individual-level controls, and β_{ct} is the country-year-specific slope for automation risk. These slopes are modeled as:

$$\beta_{ct} = \gamma_{10} + u_{1c} + v_{1t} + \omega_{ct}, \quad (2)$$

where γ_{10} is the global baseline slope, u_{1c} and v_{1t} are country and year deviations, and ω_{ct} is the country-year interaction. ε_{ict} is individual-level error clustered by country-year. We estimate this model once per exposure index on the full labor-force sample from 2002 through 2022, using design weights and standardizing exposure on the pooled estimation sample.

4 Replication Extension: Global Effects and Heterogeneity

The harmonized ESS panel reproduces the Frey–Osborne association with redistribution in the original 2002–2012 window. We use the benchmark outcome, labor-force restriction, relative-income control, and exposure indices. When the panel is extended through 2022, the aggregate association is smaller. The RTI results are reported in the appendix.

Table 1 summarizes the fitted Frey–Osborne slope by time window. To compare it with the published coefficient, we rescale each estimate to the probability’s 0–1 range. In 2002–2012, the aggregate is 0.175 (95% CI [0.098, 0.251]), which brackets the published benchmark of 0.162. In 2014–2022 it is 0.040 and not distinguishable from zero; the full-panel estimate lies between those values.

The benchmark-roster rows repeat the calculation for the seventeen countries in the published sample. They show the same pattern: the 2002–2012 aggregate is 0.147 (CI [0.069, 0.226]), while the 2014–2022 aggregate is 0.022 and again indistinguishable from zero. Thus, neither replication nor attenuation depends on the broader country coverage of the panel. The final row gives the standardized per-SD aggregate used in the figures. Taken together, these estimates point to historical attenuation alongside national variation rather than a uniform disappearance of the relationship.

Table 1: Global FO automation–redistribution association: published benchmark and post-estimation window aggregates of the single heterogeneity model

Quantity	Years	Scale	Estimate	95% CI	N	Countries
Published benchmark	2002–2012	native	0.162	–	64,639	17
Window aggregate of $\beta_{c,t}$	2002–2012	native	0.175	[0.098, 0.251]	104,889	36
Window aggregate of $\beta_{c,t}$	2014–2022	native	0.040	[-0.037, 0.117]	91,749	34
Full-panel aggregate of $\beta_{c,t}$	2002–2022	native	0.112	[0.038, 0.186]	196,638	39
Benchmark-roster aggregate	2002–2012	native	0.147	[0.069, 0.226]	69,788	17
Benchmark-roster aggregate	2014–2022	native	0.022	[-0.057, 0.102]	56,378	16
Benchmark-roster aggregate	2002–2022	native	0.091	[0.016, 0.167]	126,166	17
Full-panel aggregate of $\beta_{c,t}$	2002–2022	per SD	0.038	[0.013, 0.063]	196,638	39

Window aggregates are equal-weighted means of the country-year slopes $\beta_{c,t}$ from the single multilevel heterogeneity model; no additional model is estimated. The model adjusts for relative equivalized household income, years of education, gender, age, union membership, religiosity, and unemployment status, with ESS design weights. Benchmark-roster rows restrict the aggregation to the seventeen countries of the published benchmark’s sample, holding country composition fixed at the benchmark roster. Native-scale entries divide the per-SD aggregate by the pooled exposure SD. Intervals carry over the fixed-slope sampling variance and the aggregated conditional variances of the slope components; cross-component posterior covariances are ignored. The published-benchmark row is quoted from Thewissen and Rueda (2019) as an external reference.

Figure 2 displays estimates for every country-year. They range from negative to strongly positive values. Country-level variation exceeds temporal variation, so the association is more stable within countries than across survey waves. This pattern is consistent with the proposition that national settings condition how occupational exposure relates to redistribution preferences; it does not by itself establish an institutional cause. The appendix provides the corresponding RTI plot.

Figure 2: Country-by-year heterogeneity in the automation–redistribution association

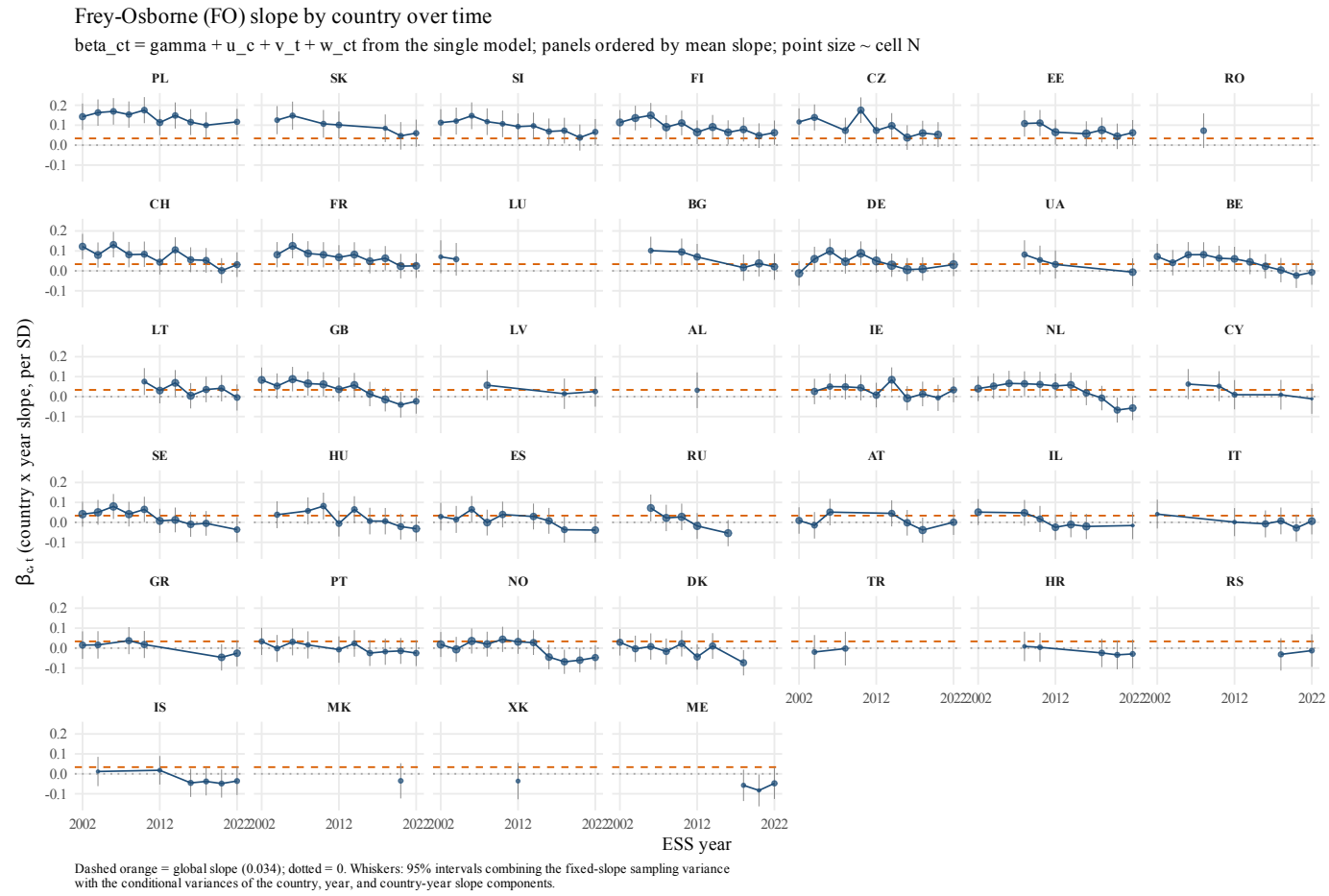


Figure 3 separates the temporal pattern from changing country coverage. It aggregates the same fitted $\beta_{c,t}$ by year for four country rosters: all observed countries, a common panel, the seventeen countries in the Thewissen and Rueda [2019] sample, and an eleven-country balanced panel. Equal country weights make differences among curves attributable to roster composition; no coefficient is re-estimated.

The four series differ in level, but each declines after 2014, reaches a low point in 2020, and partly recovers in 2022. The pattern therefore does not appear to be driven by changing country coverage. The 95% intervals combine the fixed-slope and year-component variance with averaged conditional variances for country and country-year components. The appendix reports the RTI counterpart.

Figure 3: Yearly Frey–Osborne automation–redistribution slopes under four composition assumptions

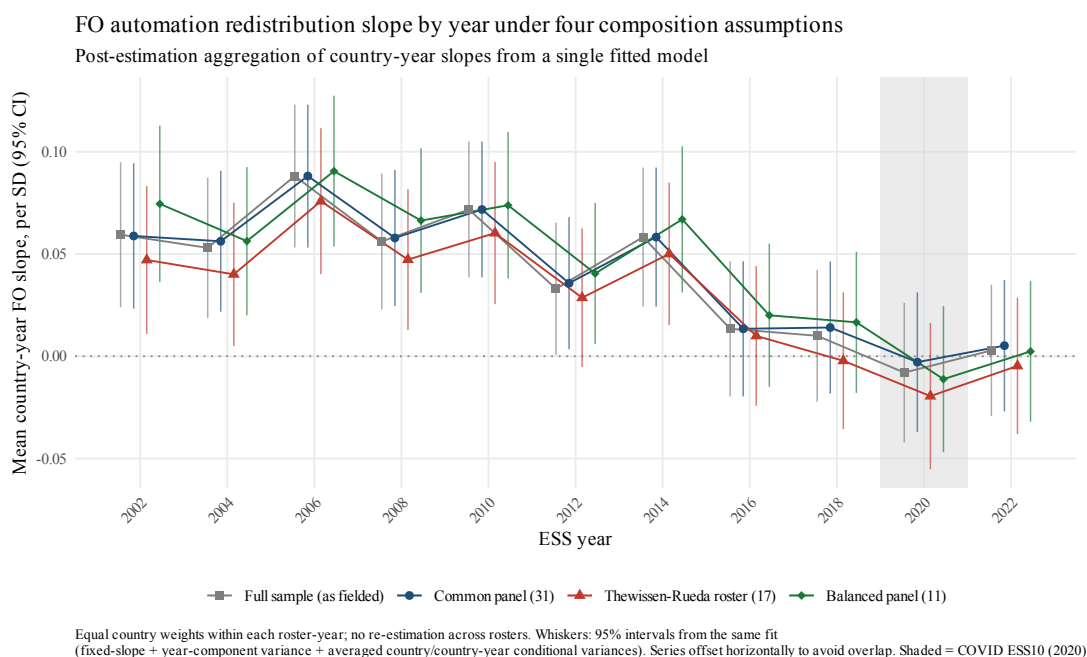
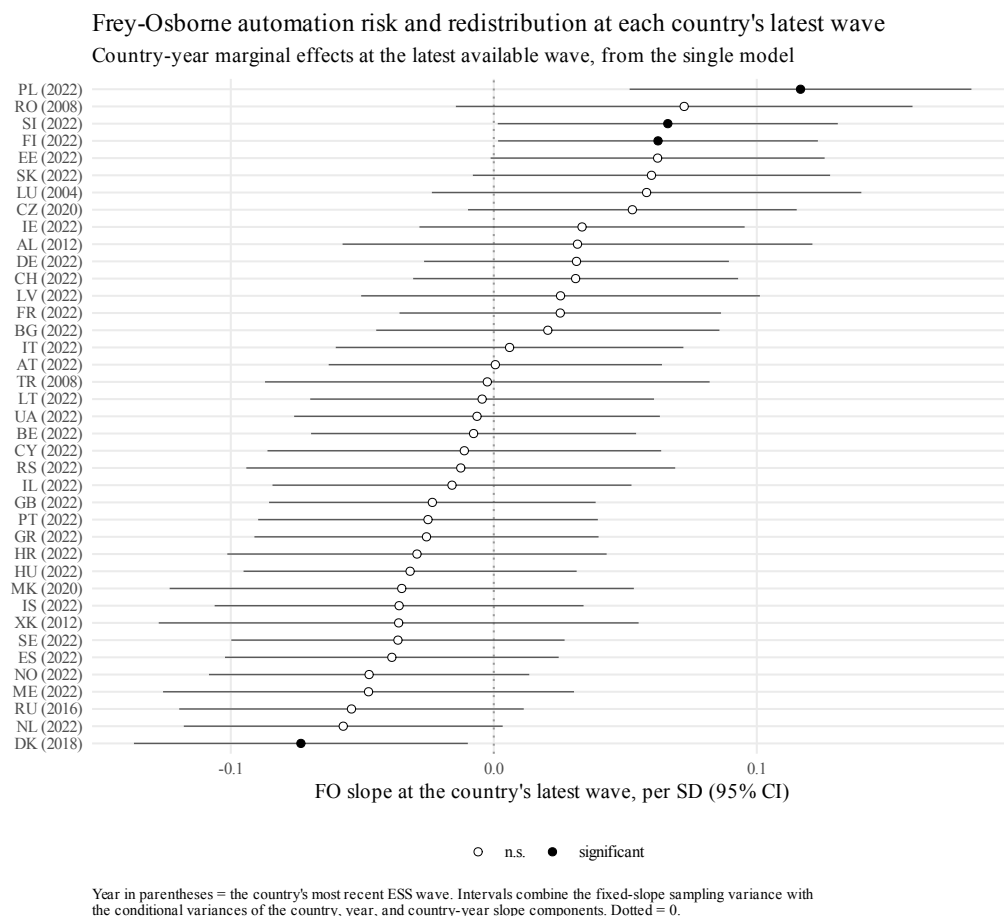


Figure 4 reports the marginal effect of Frey–Osborne exposure in the latest available observation for each country. The figure labels the year because nine of the 39 countries were last observed before 2022. Most associations are not statistically distinguishable from zero. Poland (0.117 per standard deviation), Slovenia (0.066), and Finland (0.062) have positive significant effects, while Denmark (−0.073) has a negative significant effect. The range, from −0.07 to 0.12 per standard deviation, shows that the post-2014 association is specific to particular national

contexts rather than a general European pattern. The appendix reports the RTI counterpart.

Figure 4: Country marginal effects of Frey–Osborne exposure at the most recent available wave



5 Mechanism Probes: Vulnerability and Institutional Moderation

The paper's main estimand is the association between automation exposure and redistribution preferences. The analyses in this section probe mechanisms behind that association. At the individual level, we ask whether automation exposure registers as perceived economic vulnerability. At the country level, we ask whether welfare-state and labor-market institutions moderate the exposure–vulnerability link, helping explain why the redistribution association is stronger in some national contexts than others.

Figure 5 presents the individual-level mechanism results. Across all eleven waves, automation exposure is positively associated with financial strain. In the rounds that include them,

exposure is also positively associated with expected job loss, expected income insufficiency, and job insecurity. These outcomes help assess whether occupational exposure is linked to the perceived vulnerability through which it may shape redistribution preferences.

Figure 5: Automation exposure and perceived economic vulnerability across ESS waves

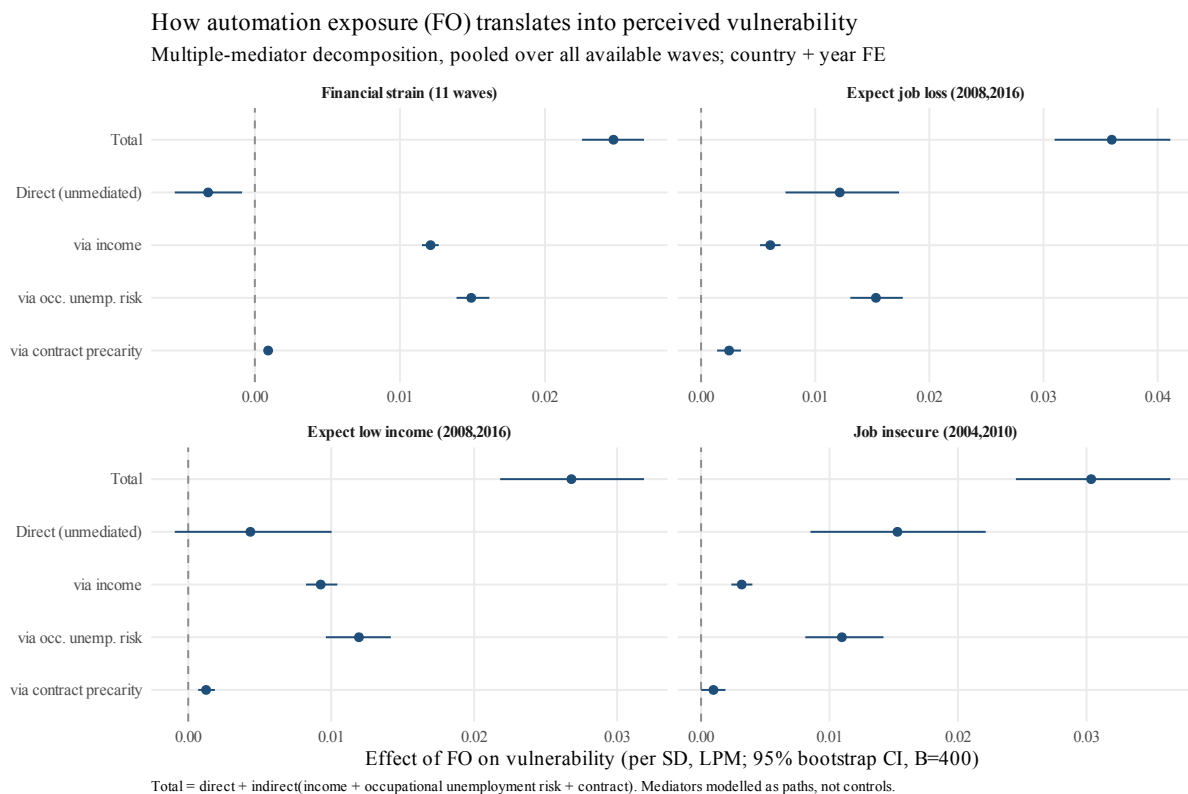
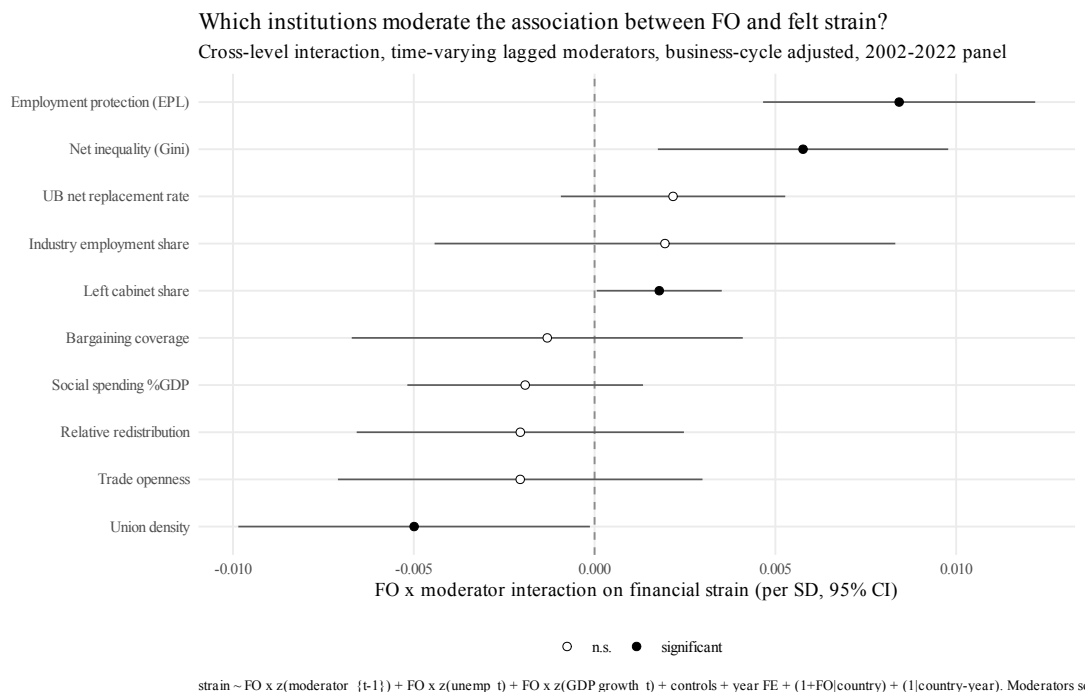


Figure 6 examines whether institutions condition the association between automation risk and financial strain. The models use one-year-lagged moderators, year fixed effects, a country-year random intercept, and business-cycle adjustment. Survey-year unemployment and real GDP growth enter as both main effects and interactions with exposure.

This adjustment matters because benefits and spending expand in downturns, when the exposure–strain gradient also steepens. Without adjustment, this countercyclical co-movement obscures a buffering pattern. Net of the cycle, within-country estimates resemble the between-country diagnostic: union density weakens the exposure–strain association for both exposure measures; bargaining coverage and social spending show the same direction for RTI; and net income inequality and employment-protection strictness strengthen the association for Frey–Osborne exposure. The last result is consistent with insider–outsider dualization, in which strict protection shelters insiders while leaving exposed outsiders more precarious. Because ten mod-

erators are tested separately with roughly thirty national clusters, these results are evidence consistent with institutional conditioning, not definitive causal estimates.

Figure 6: Institutional moderation of the automation–financial-strain association



Additional mechanism diagnostics, robustness checks, and full auxiliary tables appear in Appendix 8.1.

6 Discussion

The findings qualify a simple, time-invariant account of the political consequences of technological change. The association between automation risk and redistribution preferences reproduces in the pre-2016 period but attenuates in the subsequent decade.

One possible interpretation is a shift in the distribution of technological risk. Routine-biased technological change in the 2000s was concentrated among clerical and manual occupations, whereas machine learning and AI may expose professionals with higher skills and incomes to technological precarity [Gallego and Kurer, 2022]. Those respondents may have weaker material incentives for conventional income redistribution and may instead favor instruments such as social investment or basic income [Busemeyer and Sahm, 2022].

The mechanism analyses indicate why the redistribution association may concentrate in particular national contexts. Automation risk is associated with perceived vulnerability, including financial strain and expected job loss. Institutional moderation becomes clearer after business-cycle adjustment: welfare effort expands countercyclically, in the same downturns that steepen the exposure–strain association. Unadjusted interactions can therefore run against buffering.

After accounting for unemployment and growth, the results are consistent with the power-resources account [Stephens, 1979]. They are also consistent with the social-insurance account [Moene and Wallerstein, 2001]. Union density weakens the exposure–strain association within and between countries. Bargaining coverage and social spending have the same direction for RTI. Net income inequality and strict employment protection strengthen the Frey–Osborne association. These estimates are observational, test one moderator at a time, and use roughly thirty national clusters. They should be read as institutional patterns consistent with theory, not identified causal effects.

Finally, several limitations of this study warrant caution. First, our operationalization of automation risk relies on occupational-level indices (Frey–Osborne probabilities and RTI), which assume task structures are uniform within broad occupations. Second, the harmonization of occupations across the ISCO-88 and ISCO-08 systems introduced potential crosswalk measurement error. Third, changing country coverage across different ESS waves prevents us from estimating a perfectly balanced panel, though we address this via composition-adjusted common-sample models. Finally, our observational design limits our ability to make causal claims, and we treat the associations as descriptive patterns filtered through institutional contexts.

7 Conclusion

Using harmonized European Social Survey data for 2002–2022, this paper finds that the association between routine-task automation risk and redistribution demand is reproducible in the original period, attenuated over time, and heterogeneous across countries. After adjustment for business-cycle dynamics, labor organization weakens the association between automation exposure and perceived vulnerability, while income inequality and insider-protecting employment

regulation strengthen it.

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8 Appendix

8.1 Supplementary Figures and Tables

This appendix reports RTI robustness results, country and year slope profiles, composition aggregates, latest-wave country estimates, mediation diagnostics, and the institutional two-stage diagnostic.

Table 2: Global RTI automation–redistribution association: published benchmark and post-estimation window aggregates of the single heterogeneity model

Quantity	Years	Scale	Estimate	95% CI	<i>N</i>	Countries
Published benchmark	2002–2012	per SD	0.042	–	64,639	17
Window aggregate of $\beta_{c,t}$	2002–2012	native	0.039	[0.020, 0.058]	98,068	36
Window aggregate of $\beta_{c,t}$	2014–2022	native	0.015	[-0.004, 0.035]	83,543	34
Full-panel aggregate of $\beta_{c,t}$	2002–2022	native	0.028	[0.010, 0.046]	181,611	39
Benchmark-roster aggregate	2002–2012	native	0.042	[0.022, 0.062]	65,134	17
Benchmark-roster aggregate	2014–2022	native	0.019	[-0.001, 0.039]	51,216	16
Benchmark-roster aggregate	2002–2022	native	0.032	[0.013, 0.050]	116,350	17
Full-panel aggregate of $\beta_{c,t}$	2002–2022	per SD	0.026	[0.009, 0.043]	181,611	39

Window aggregates are equal-weighted means of the country-year slopes $\beta_{c,t}$ from the single multilevel heterogeneity model; no additional model is estimated. The model adjusts for relative equivalized household income, years of education, gender, age, union membership, religiosity, and unemployment status, with ESS design weights. Benchmark-roster rows restrict the aggregation to the seventeen countries of the published benchmark’s sample, holding country composition fixed at the benchmark roster. Native-scale entries divide the per-SD aggregate by the pooled exposure SD. Intervals carry over the fixed-slope sampling variance and the aggregated conditional variances of the slope components; cross-component posterior covariances are ignored. The published-benchmark row is quoted from Thewissen and Rueda (2019) as an external reference.

Figure 7: Marginal country slopes from the single heterogeneity model

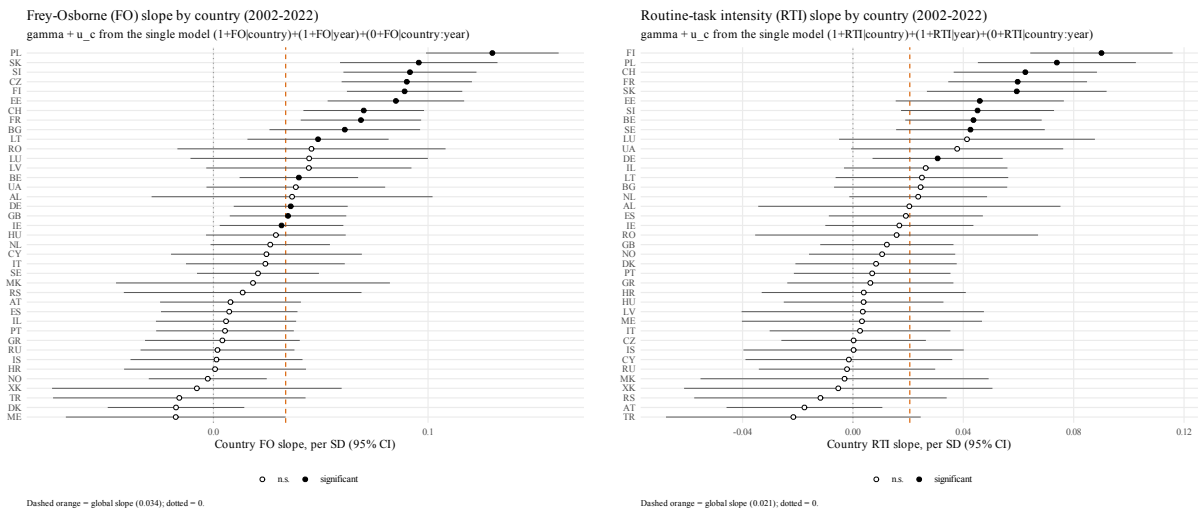


Figure 8: RTI robustness: country-by-year heterogeneity in the automation–redistribution association

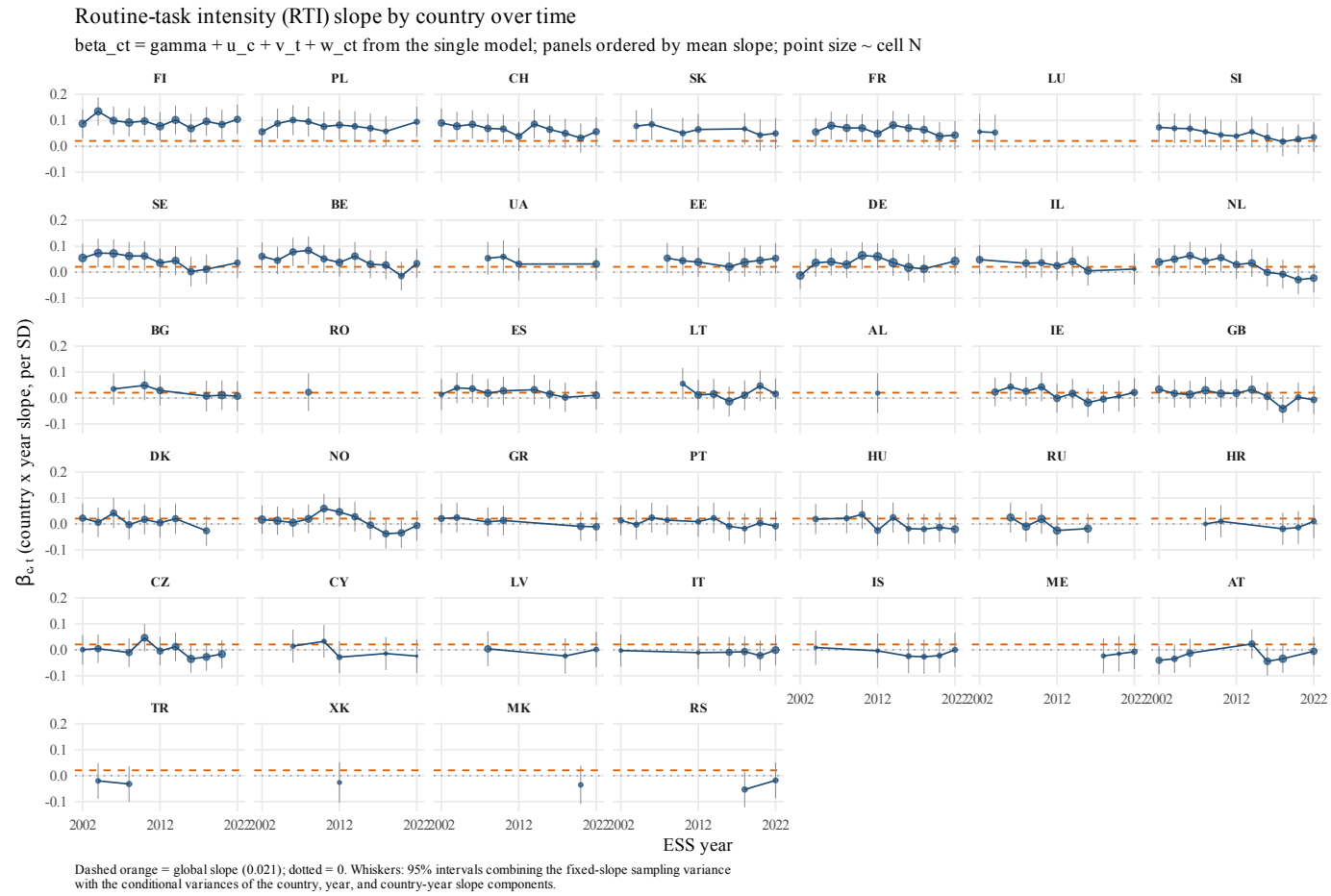


Figure 9: Marginal year slopes from the single heterogeneity model (composition-confounded)

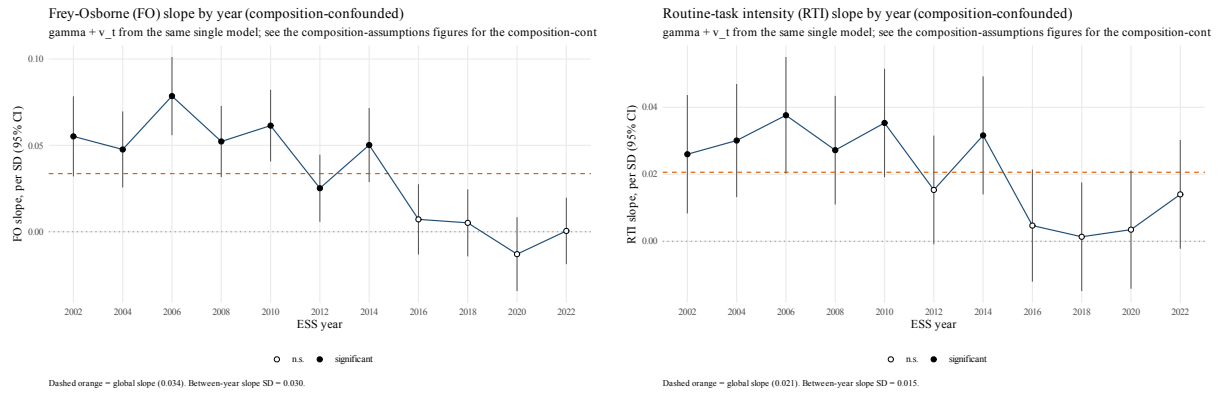


Figure 10 gives the RTI counterpart to Figure 3. It aggregates the fitted RTI country-year slopes $\beta_{c,t}$ over the same four rosters and constructs intervals in the same way. The RTI series also differ in level across rosters, decline after 2014, and partly recover in 2022.

Figure 10: RTI robustness: yearly mean country-year slopes under four composition assumptions

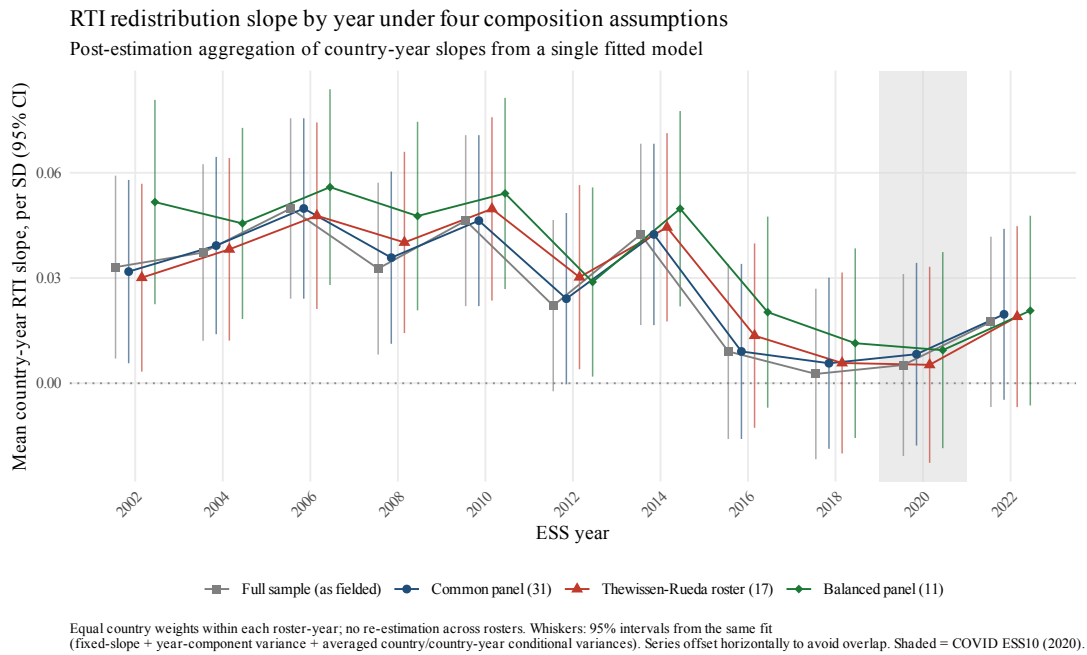


Figure 11: RTI robustness: country marginal effects at the most recent available wave

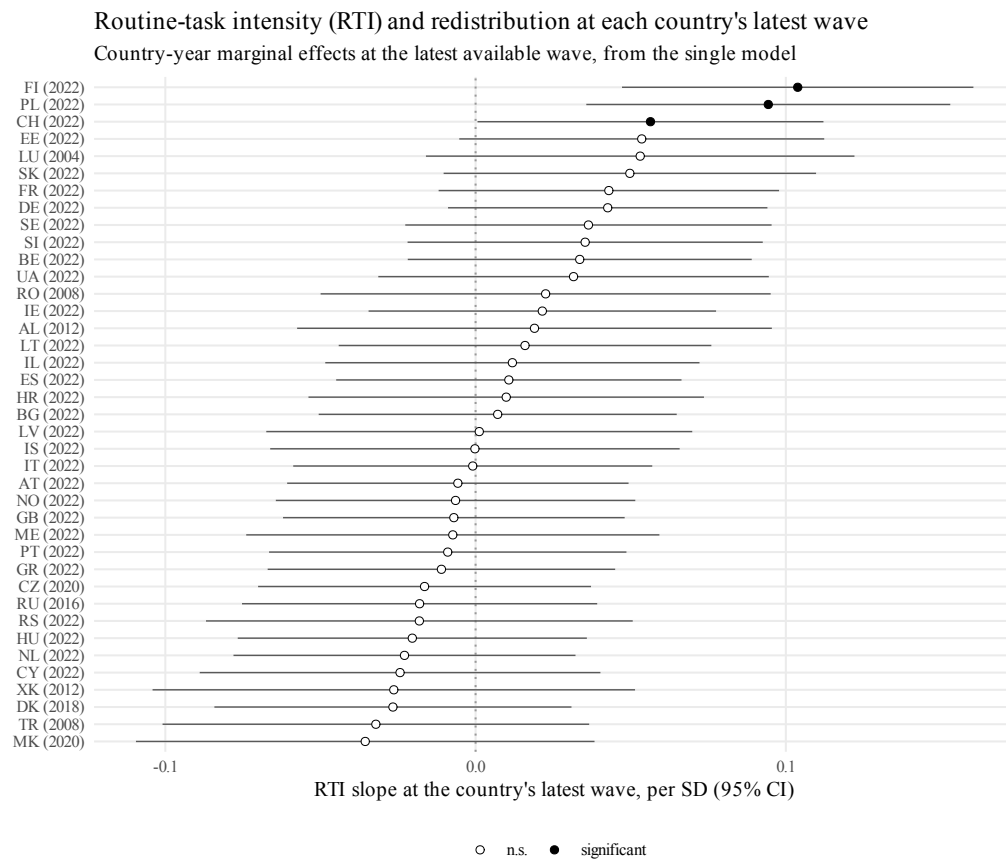


Figure 12: Country heterogeneity in automation exposure and financial strain

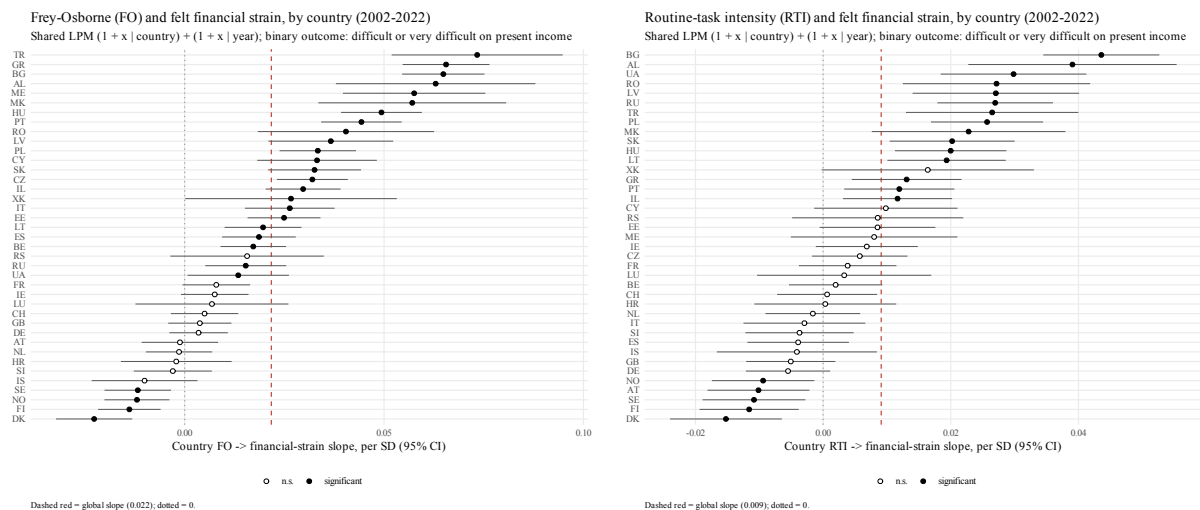


Figure 13: Mediation decomposition and path diagrams (2016 wave)

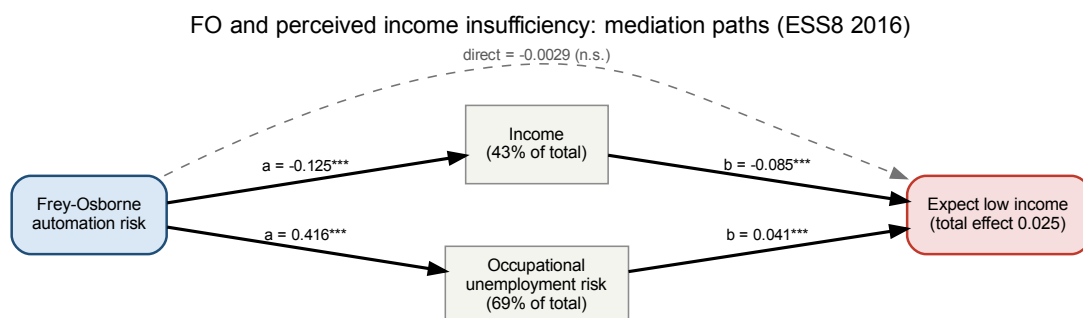
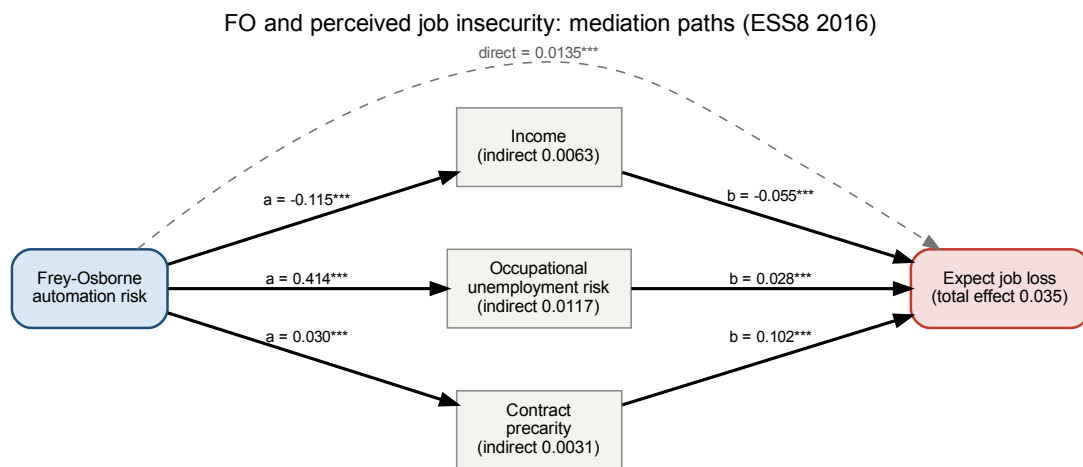
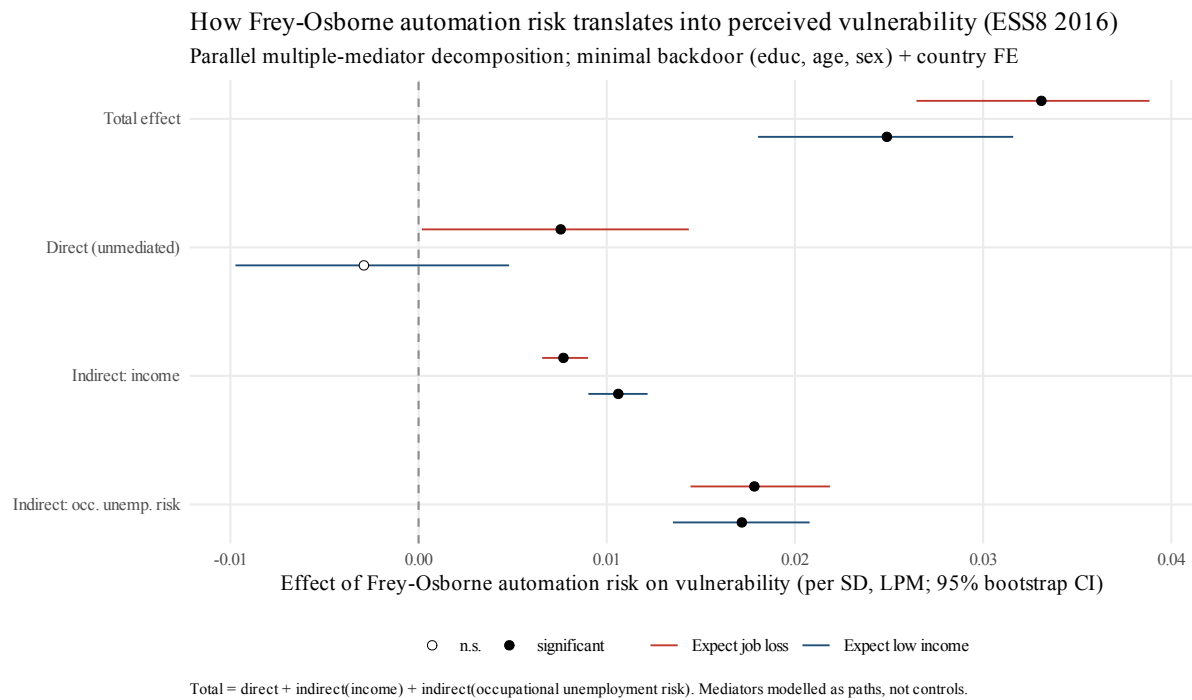
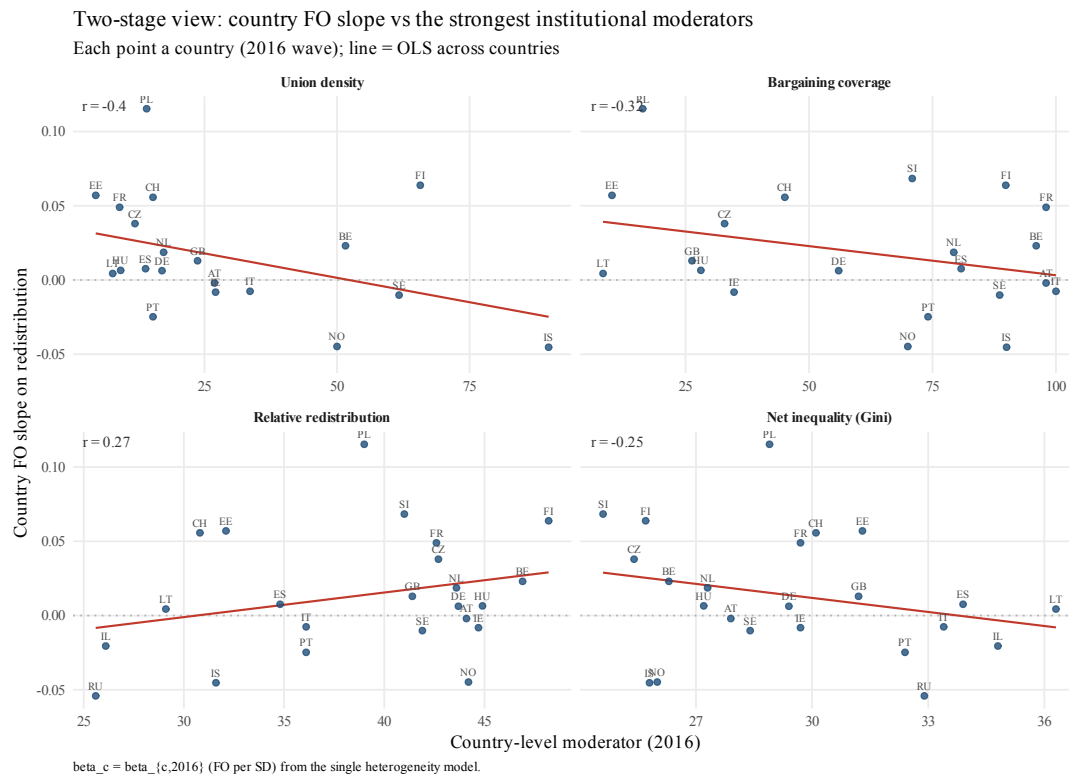


Figure 14: Two-stage institutional diagnostic for redistribution heterogeneity



8.2 NACE Rev. 2.1: First-Level Economic-Activity Classification

Table 3: Statistical Classification of Economic Activities in the European Community, Rev. 2.1 (NACE Rev. 2.1)

Section	Code Range	Description
A	01:03	Agriculture, Forestry and Fishing
B	05:09	Mining and Quarrying
C	10:33	Manufacturing
D	35:35	Electricity, Gas, Steam and Air Conditioning Supply
E	36:39	Water Supply; Sewerage, Waste Management and Remediation Activities
F	41:43	Construction
G	45:47	Wholesale and Retail Trade
H	49:53	Transportation and Storage
I	55:56	Accommodation and Food Service Activities
J	58:60	Publishing, Broadcasting, and Content Production and Distribution Activities
K	61:63	Telecommunication, Computer Programming, Consulting, Computing Infrastructure and Other Information Service Activities
L	64:66	Financial and Insurance Activities
M	68:68	Real Estate Activities
N	69:75	Professional, Scientific and Technical Activities
O	77:82	Administrative and Support Service Activities
P	84:84	Public Administration and Defence; Compulsory Social Security
Q	85:85	Education
R	86:88	Human Health and Social Work Activities
S	90:93	Arts, Sports and Recreation
T	94:96	Other Service Activities
U	97:98	Activities of Households as Employers and Undifferentiated Goods - and Service-Producing Activities of Households for Own Use
V	99:99	Activities of Extraterritorial Organisations and Bodies

8.3 Standard Occupational Classification (SOC) System

The 2010 Standard Occupational Classification (SOC-2010) is the U.S. statistical system for classifying occupations. Federal agencies use it to organize occupational information and labor-market data.

SOC-2010 is hierarchical, moving from broad groups to specific occupations identified by unique codes. It contains more than 800 detailed occupation codes.

Major Categories of SOC-2010

The table below lists the SOC-2010 major groups.

Table 4: Major Categories of SOC-2010

Code	Major Group	Code	Major Group
11	Management Occupations	33	Protective Service Occupations
13	Business and Financial Operations Occupations	35	Food Preparation and Serving Related Occupations
15	Computer and Mathematical Occupations	37	Building and Grounds Cleaning and Maintenance Occupations
17	Architecture and Engineering Occupations	39	Personal Care and Service Occupations
19	Life, Physical, and Social Science Occupations	41	Sales and Related Occupations
21	Community and Social Service Occupations	43	Office and Administrative Support Occupations
23	Legal Occupations	45	Farming, Fishing, and Forestry Occupations
25	Education, Training, and Library Occupations	47	Construction and Extraction Occupations
27	Arts, Design, Entertainment, Sports, and Media Occupations	49	Installation, Maintenance, and Repair Occupations
29	Healthcare Practitioners and Technical Occupations	51	Production Occupations
31	Healthcare Support Occupations	53	Transportation and Material Moving Occupations

Table 5: Descriptive statistics for the analysis variables of the harmonised 2002–2022 labour-force panel (all eleven ESS waves). Each row is a model variable; the columns report the number of non-missing observations (N), the mean and standard deviation, and the minimum, the tenth, twenty-fifth, fiftieth, seventy-fifth, and ninetieth percentiles, and the maximum. Binary indicators have means equal to their prevalence in the sample. The Frey–Osborne automation probability is on its native 0–1 scale, the routine-task-intensity index on its native GMS scale, and relative equivalized income enters as the within-country-year z-score. The vulnerability items exist only in the waves noted.

[illegible]

8.5 ESS Variable Dictionary

Table 6: ESS variable dictionary

Variable	Description	Value labels
y	Redistribution preference; the ESS item gincdif reverse-coded ($y = 6 - \text{gincdif}$) so higher values mean stronger support for the government reducing income differences.	'5' = 'Agree strongly'; '4' = 'Agree'; '3' = 'Neither agree nor disagree'; '2' = 'Disagree'; '1' = 'Disagree strongly'
fo	Frey-Osborne (2017) probability that the respondent's occupation is automatable, matched to the occupation code and harmonized across the ISCO-88 and ISCO-08 waves.	Continuous, 0 to 1
rti	Goos-Manning-Salomons routine-task-intensity of the occupation (appendix robustness exposure), harmonized across the ISCO seam.	Continuous; higher is more routine-task-intensive
income	Thewissen-Rueda relative equivalized household income, entered as the within country-year z-score. Built from the ESS show-card income band midpoint (Pareto mean for the open top band), square-root household equivalization, and the ratio to the design-weighted country-year mean.	Continuous z-score
eduyrs	Years of full-time education completed (ESS eduyrs).	Continuous, years
male	Respondent gender, from ESS gndr.	'1' = 'Male'; '0' = 'Female'
age	Age of the respondent in years (ESS agea).	Continuous, years
union	Trade-union membership, from ESS mbtru.	'1' = 'Member, current or former'; '0' = 'Never a member'
relig	Subjective religiosity on the ESS rlgdgr scale.	'0' = 'Not at all religious'; '10' = 'Very religious'
unemp	Unemployed under the Thewissen-Rueda labour-force definition (ESS mnactic coded as unemployed, looking or not looking for work).	'1' = 'Unemployed'; '0' = 'Employed'

Continued on next page

Variable	Description	Value labels
occ_risk	Occupational unemployment risk, Rehm-style: the design-weighted, leave-one-out unemployment rate by wave and two-digit ISCO under the ILO labour-force definition (minimum cell size of 30).	Continuous, share unemployed
no_or_limtd_contract	Contract precarity, from ESS wrkctra.	'1' = 'Limited-duration or no contract'; '0' = 'Unlimited contract'
d_fin_strain	Subjective financial strain, from ESS hincfel; available in all eleven waves.	'1' = 'Difficult or very difficult on present income'; '0' = 'Coping or living comfortably'
d_job_vulnr	Expected job loss, from ESS lkuemp (likely or very likely to become unemployed in the next twelve months). Welfare module, 2008 and 2016.	'1' = 'Likely or very likely'; '0' = 'Not or not very likely'
d_income_vulnr	Expected income insufficiency, from ESS lknemny (likely or very likely to have insufficient household income in the next twelve months). Welfare module, 2008 and 2016.	'1' = 'Likely or very likely'; '0' = 'Not or not very likely'
d_job_insecure	Job insecurity, from ESS jbscr (rates own job security low). Work module, 2004 and 2010.	'1' = 'Insecure'; '0' = 'Secure'
dweight	ESS design weight, applied in all estimation.	Continuous weight
country	ESS country of interview (39 countries, ISO codes).	Categorical
year	ESS survey year and wave, 2002 to 2022 (rounds 1 to 11).	Categorical, 2002 to 2022